

## Logical operators binary truth table definition

| Input |     | Output                      |                              |                           |                                  |                                        |
|-------|-----|-----------------------------|------------------------------|---------------------------|----------------------------------|----------------------------------------|
| $p$   | $q$ | Conjunction<br>$p \wedge q$ | Exclusive or<br>$p \oplus q$ | Disjunction<br>$p \vee q$ | Conditional<br>$p \rightarrow q$ | Biconditional<br>$p \leftrightarrow q$ |
| $T$   | $T$ | $T$                         | $F$                          | $T$                       | $T$                              | $T$                                    |
| $T$   | $F$ | $F$                         | $T$                          | $T$                       | $F$                              | $F$                                    |
| $F$   | $T$ | $F$                         | $T$                          | $T$                       | $T$                              | $F$                                    |
| $F$   | $F$ | $F$                         | $F$                          | $F$                       | $T$                              | $T$                                    |
|       |     | “ $p$ and $q$ ”             | “ $p$ xor $q$ ”              | “ $p$ or $q$ ”            | “if $p$ then $q$ ”               | “ $p$ if and only if $q$ ”             |

## Logical operators truth tables

Truth tables: Input-output tables where we use  $T$  for 1 and  $F$  for 0.

| Input |     | Output                                                                              |                                                                                     |                                                                                     |
|-------|-----|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| $p$   | $q$ | Conjunction<br>$p \wedge q$                                                         | Exclusive or<br>$p \oplus q$                                                        | Disjunction<br>$p \vee q$                                                           |
| $T$   | $T$ | $T$                                                                                 | $F$                                                                                 | $T$                                                                                 |
| $T$   | $F$ | $F$                                                                                 | $T$                                                                                 | $T$                                                                                 |
| $F$   | $T$ | $F$                                                                                 | $T$                                                                                 | $T$                                                                                 |
| $F$   | $F$ | $F$                                                                                 | $F$                                                                                 | $F$                                                                                 |
|       |     |  |  |  |

| Input                                                                                 | Output   |
|---------------------------------------------------------------------------------------|----------|
| $p$                                                                                   | $\neg p$ |
| $T$                                                                                   | $F$      |
| $F$                                                                                   | $T$      |
|  |          |